DAY 4

Other Protocols in Action

FTP and SNMP – Theory, Configuration and Practical Exercises

Duration: 1 Hour | **Theory:** 30 Minutes | **Practical:** 30 Minutes

Session overview

This training material introduces File Transfer Protocol (FTP) and Simple Network Management Protocol (SNMP), then guides students through configuring an FTP server in Cisco Packet Tracer and transferring files between devices.

1. Learning Objectives

• Understand FTP and SNMP protocols.

• Explain how FTP transfers files between a client and a server.

• Understand the role of SNMP in network monitoring.

• Identify FTP and SNMP port numbers.

• Configure an FTP server in Cisco Packet Tracer.

• Transfer files between a client and an FTP server.

• Compare FTP and HTTP based on their purpose and functionality.

PART A: THEORY (30 MINUTES)

2. What is FTP?

FTP stands for **File Transfer Protocol**. FTP is a network protocol used to transfer files between a client computer and a server over a network. It allows users to upload, download, and manage files stored on a remote server.

Real-time example: A web developer creates a website on a computer. The website contains HTML, CSS, JavaScript, images, and documents. The developer uses FTP to upload these files to a hosting server. Files can also be downloaded from the server to the local computer.

3. Why Do We Need FTP?

Imagine a company has a central file server where employees store documents. Instead of sending large files through email or messaging applications, employees can use a file transfer service to upload and download files.

• Transfers files between computers.

• Supports uploading and downloading.

• Helps manage remote server files.

• Supports transferring multiple files.

• Useful for website deployment and file sharing.

4. How Does FTP Work?

FTP follows a client-server architecture. The FTP client is the computer or application that connects to the server and requests file operations. The FTP server stores files and allows authorized clients to upload or download them.

FTP Client	**FTP Communication**	**FTP Server**
User's computer	Login, upload, download and file management Control connection: TCP port 21	Stores and manages files

FTP Working Process

• **Step 1 – Connection:** The FTP client connects to the FTP server using its IP address or hostname.

• **Step 2 – Authentication:** The client provides a username and password if authentication is required.

• **Step 3 – File Selection:** The user selects the file to upload or download.

• **Step 4 – File Transfer:** The FTP client transfers the selected file to or from the server.

• **Step 5 – Connection Termination:** After completing the transfer, the user disconnects from the FTP server.

5. FTP Ports

Connection	Port	Purpose
Control connection	TCP 21	Login and FTP commands
Data connection	TCP 20 in active mode	File and directory data in active mode

In passive FTP, the server negotiates a separate data port instead of using TCP port 20.

Security note: Standard FTP does not encrypt usernames, passwords, or transferred files. Secure alternatives include SFTP (SSH File Transfer Protocol, commonly TCP port 22) and FTPS (FTP secured using TLS). SFTP and FTPS are different protocols.

6. Common FTP Commands

Command	Purpose
open	Connect to an FTP server
user	Provide username
ls	Display files and directories
get	Download a file
put	Upload a file
delete	Delete a remote file
mkdir	Create a directory
bye	Disconnect

Example: A developer uses `put index.html` to upload a website's homepage to an FTP server.

7. What is SNMP?

SNMP stands for **Simple Network Management Protocol**. SNMP is used to monitor and manage network devices such as routers, switches, servers, printers, and access points. It allows network administrators to collect information from devices through a central monitoring system.

Real-time example: A company has 20 switches, 10 routers, and multiple servers. A network administrator wants to monitor which devices are online, bandwidth usage, interface status, CPU and memory usage (if supported), and device alerts. SNMP helps collect this information.

8. How Does SNMP Work?

SNMP mainly uses two components: the **SNMP Manager**, which is the central monitoring system, and the **SNMP Agent**, which runs on a network device and responds to management requests. Management information is organized using a Management Information Base (MIB).

SNMP Manager Central monitoring system	**SNMP communication** GET / RESPONSE / SET / TRAP UDP ports 161 and 162	**SNMP Agents** Router / Switch / Server

SNMP Working Process

• **Step 1 – Request:** The SNMP Manager sends a request to a network device to collect information.

• **Step 2 – Agent:** The SNMP Agent receives the request and collects the requested information.

• **Step 3 – Response:** The agent sends the information back to the SNMP Manager.

• **Step 4 – Monitoring:** The administrator views device information through a monitoring dashboard.

• **Step 5 – Alert:** A device can send an SNMP Trap when an important event occurs.

9. Important SNMP Terms

Term	Meaning
Manager	Central monitoring system
Agent	Software running on the network device
MIB	Database structure containing management information
OID	Object Identifier used to identify specific information
GET	Requests information from a device
SET	Changes a supported configuration value
TRAP	Sends an unsolicited event notification
INFORM	Sends a notification that expects acknowledgement

10. SNMP Versions and Ports

Version	Description
SNMPv1	Original version with basic monitoring.
SNMPv2c	Improved performance and community-string-based access.
SNMPv3	Supports authentication, integrity, and encryption.

• **UDP 161:** Requests and responses.

• **UDP 162:** Traps and notifications.

Security note: SNMPv1 and SNMPv2c community strings are not encrypted. SNMPv3 is recommended when properly configured because it supports stronger security.

11. Difference Between FTP and SNMP

Feature	FTP	SNMP
Full form	File Transfer Protocol	Simple Network Management Protocol
Main purpose	File transfer	Network monitoring and management
Architecture	Client-server	Manager-agent
Transport	TCP	UDP
Common ports	TCP 21; separate data connection	UDP 161 and 162
Main function	Upload and download files	Collect device information and manage devices
Example	Upload website files	Monitor router bandwidth

Feature	FTP	SNMP
Security	Standard FTP is unencrypted	SNMPv3 supports authentication and encryption

PART B: PRACTICAL (30 MINUTES)

Practical: Configure an FTP Server in Cisco Packet Tracer

Objective: Configure an FTP server, create user credentials, connect client computers, and transfer files between the client and server.

Requirements

• 1 Server (Server-PT)

• 1 Switch (2960)

• 2 PCs (PC0 and PC1)

• Cisco Packet Tracer

Step 1: Create Network Topology

Add the server, switch, and two PCs. Connect all devices using Copper Straight-Through cables. Connect the server and both PCs to different switch ports.

Device	IP address	Subnet mask
FTP Server	192.168.10.2	255.255.255.0
PC0	192.168.10.10	255.255.255.0
PC1	192.168.10.11	255.255.255.0

Step 2: Configure FTP Server IP Address

Click **Server** → **Desktop** → **IP Configuration** and enter:

```
IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: Leave blank
```

Step 3: Enable FTP Service

• Click Server.

• Open the Services tab.

• Select FTP.

• Turn FTP Service ON.

Step 4: Create FTP User Account

Field	Value
Username	student
Password	student123
Permissions	Read, Write, Delete, Rename, List

Click **Add**. For this lab, all permissions are enabled to demonstrate FTP operations. In real environments, grant only the permissions each user needs.

Step 5: Configure PC0

Click PC0 → Desktop → IP Configuration and enter:

```
IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Default Gateway: Leave blank
```

Step 6: Configure PC1

Click PC1 → Desktop → IP Configuration and enter:

```
IP Address: 192.168.10.11
Subnet Mask: 255.255.255.0
Default Gateway: Leave blank
```

Step 7: Test FTP Server Connectivity

On PC0, open Desktop → Command Prompt and execute:

```
ping 192.168.10.2
```

Step 8: Connect to FTP Server

On PC0, open Command Prompt and enter:

```
ftp 192.168.10.2
```

When prompted, enter the username and password:

```
Username: student
Password: student123
```

You should now be connected to the FTP server.

Step 9: View Files on FTP Server

```
dir
help
```

The `dir` command displays available files. The `help` command displays supported FTP commands.

Step 10: Download a File

Ensure a sample file named `student_notes.txt` exists on the FTP server. Then execute:

```
get student_notes.txt
```

The file will be downloaded from the FTP server to PC0.

Step 11: Upload a File

On PC0, open Desktop → Text Editor. Create a file named `my_report.txt`, add sample text, and save it. Return to the FTP command prompt and execute:

```
put my_report.txt
```

The file will be uploaded from PC0 to the FTP server. If file creation or upload is unsupported in your Packet Tracer version, use an available local file to test the transfer.

Step 12: Verify Uploaded Files

```
dir
bye
```

Use `dir` to verify the uploaded file and `bye` to disconnect.

Expected result: The FTP server is configured, the user account is created, PC0 connects successfully, a file is downloaded, a file is uploaded, and the uploaded file is visible in the server file list.

PART C: ADDITIONAL PRACTICAL CHALLENGES

Challenge 1: Multiple Client Connections

Connect PC1 to the same FTP server. On PC1, open Command Prompt and execute:

```
ftp 192.168.10.2
```

Log in using the same credentials and ask students to download the available files. Learning point: multiple client computers can access a central FTP server, subject to its permissions and connection limits.

Challenge 2: FTP Permission Testing

On Server → Services → FTP, create another user with the following settings:

Field	Value
Username	readonly
Password	read123
Read	Enabled
Write	Disabled
Delete	Disabled
Rename	Disabled
List	Enabled

Connect from PC1 and log in with the readonly account:

```
ftp 192.168.10.2
Username: readonly
Password: read123
```

Test downloading and uploading:

```
get student_notes.txt
put my_report.txt
```

Expected observation: Download should be allowed. Upload should be denied because Write permission is disabled. This demonstrates how FTP permissions control file access.

PART D: STUDENT ASSIGNMENT

Task: Create a Comparison Table Between FTP and HTTP

Students must prepare a comparison table explaining the differences between FTP and HTTP.

Feature	FTP	HTTP
Full form	File Transfer Protocol	Hypertext Transfer Protocol
Purpose	Transfers files between client and server	Transfers web content between browser and server
Main use	Uploading and downloading files	Accessing websites and web applications
Architecture	Client-server	Client-server
Transport	TCP	TCP traditionally; HTTP/3 uses QUIC over UDP
Default port	TCP 21 (control connection)	TCP 80
Secure version	FTPS / SFTP	HTTPS
Secure port	FTPS commonly TCP 990 for implicit FTPS; SFTP TCP 22	HTTPS commonly TCP 443
Authentication	Can use username and password	Depends on website or application

Feature	FTP	HTTP
Example	Uploading website files to hosting	Opening a website in Chrome

Real-time Examples

FTP: A developer uploads index.html, style.css, and images to a hosting server using FTP.

HTTP: A user opens a website in Chrome. The browser sends an HTTP request, and the web server returns the requested webpage.

Assignment Questions

• What is FTP?

• What is SNMP?

• Explain FTP client-server architecture.

• What is the difference between FTP and SFTP?

• What are the control and data connections in FTP?

• What is the role of an SNMP Manager?

• What is an SNMP Agent?

• What is MIB?

• Which ports are used by FTP and SNMP?

• Why is SNMPv3 more secure than SNMPv1 and SNMPv2c?

• Explain the difference between FTP and HTTP with real-time examples.

Submission Requirements

• Packet Tracer topology screenshot.

• FTP service configuration screenshot.

• FTP user account screenshot.

• Successful FTP login screenshot.

• File download screenshot.

• File upload screenshot.

• FTP vs HTTP comparison table.

• Answers to assignment questions.

Save the Packet Tracer file as: `Day4_FTP_Configuration.pkt`

PART E: END-OF-DAY QUICK REVISION

Topic	Quick Answer
FTP full form	File Transfer Protocol
FTP purpose	File transfer
FTP control port	TCP 21
FTP active data port	TCP 20
Secure FTP alternatives	SFTP / FTPS

Topic	Quick Answer
SNMP full form	Simple Network Management Protocol
SNMP purpose	Network monitoring and management
SNMP Manager	Central monitoring system
SNMP Agent	Runs on network devices
SNMP MIB	Management Information Base
SNMP request/response port	UDP 161
SNMP trap port	UDP 162
FTP upload command	put
FTP download command	get
FTP file listing	dir

Trainer's Practical Discussion

• If a company wants to upload website files to its hosting server, which protocol can it use?

• If an employee wants to download a document from a remote file server, which protocol can help?

• If a network administrator wants to monitor 50 switches from one dashboard, which protocol can help?

• If a switch suddenly goes offline, how can SNMP help the administrator identify the issue?

• Why should companies avoid using standard FTP for transferring confidential files?

Expected Learning Outcome: Students should understand FTP and SNMP, configure an FTP server in Cisco Packet Tracer, transfer files between client and server, understand FTP permissions, and compare FTP with HTTP.